Seznu Tabussum Jesun

AI / Machine Learning Engineer | Medical Image Analysis & Computer Vision
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PROFESSIONAL SUMMARY

Final-year Computer Science & Engineering student specializing in deep learning for medical image segmentation and classification. Designed PRISM-Net, a lightweight polyp segmentation architecture achieving 0.9109 Dice on Kvasir-SEG with validated cross-dataset generalization. Experienced in the full research pipeline: model design, ablation studies, multi-seed evaluation, robustness testing, and IEEE-format publication writing. Seeking an AI/ML role to build reliable, efficient models for real-world problems.

EDUCATION

B.Sc. in Computer Science and Engineering Final Year
International Islamic University Chittagong (IIUC), Chattogram, Bangladesh
- Thesis: PRISM-Net: Pyramid Residual Integration and Separable Multi-scale U-Net for Polyp Segmentation

RESEARCH & PROJECTS

PRISM-Net: Polyp Segmentation (Final Year Thesis) Kvasir-SEG, CVC-ClinicDB, CVC-ColonDB, ETIS-LaribPolypDB
- Designed a U-Net variant combining a pretrained ResNet34 encoder, ASPP bottleneck, attention-gated skip connections, and a depthwise-separable decoder.
- Achieved Dice 0.9109 and IoU 0.8551 on Kvasir-SEG, reported with confidence intervals over six random seeds.
- Evaluated cross-dataset generalization on three unseen colonoscopy datasets; ran ablations on architecture components, loss functions (Focal Tversky, boundary-aware), and threshold optimization.
- Applied discriminative learning rates and mixed-precision training; wrote thesis chapters in LaTeX.

STAR: PM2.5 Forecasting in Bangladesh Time-series, 16 stations, 30,095 station-days
- Proposed a Station-aware Temporal-Attention Residual network: a single Transformer self-attention layer over a 14-day window predicts Prophet residuals, with FiLM layers conditioning on the monitoring station.
- Achieved the lowest MAE (**16.71 vs 17.15** for Prophet+LSTM) and best symmetric MAPE; significantly better than the LSTM per-sample ($p<10^{-5}$, Holm-corrected) at lower architectural cost.
- Ran a rigorous evaluation: leakage-safe per-station split, bootstrap CIs, Friedman test, Holm-corrected Wilcoxon, Nemenyi analysis.

MBLNet (DSUNetV2): Lightweight Segmentation Model Efficiency-focused
- Built a MobileNetV2-style inverted-residual, depthwise-separable segmentation network from scratch with full ablation and multi-seed evaluation, matching the PRISM-Net protocol.
- Resolved CUDA out-of-memory bottlenecks via mixed-precision training and batch-size tuning.

Semi-Supervised Cervical Cytology Classification SIPaKMeD
- Implemented DA-FreeMatch with an EfficientNetV2-S backbone, adding Distribution Alignment and a KL-divergence fairness loss to handle limited labels.
- Authored a full IEEE-format paper with architecture diagrams.

Polyp Size Classification (Two-Stage Cascade) Multi-dataset
- Built a MiniVGGNet-based cascade classifier; integrated Kvasir, Kvasir-SEG, CVC-ClinicDB, CP-CHILD, and ERCPMP data with video-frame preprocessing and size-metadata parsing.
- Assessed cross-dataset generalization across heterogeneous clinical sources.

MedCNN: Adversarial Robustness on Chest X-rays Security / Robustness
- Implemented and evaluated FGSM, PGD, DeepFool, and adversarial patch attacks with corresponding defenses; prepared a conference submission.

Multimodal Bangla Disaster Severity Classification Kaggle Competition
- Iterated multimodal (image + text) models combining CLIP, XLM-RoBERTa, Swin, and DistilBERT.
- Mitigated overfitting and class imbalance using Focal Loss, WeightedRandomSampler, per-class threshold optimization, and test-time augmentation.

TECHNICAL SKILLS

Languages & Tools: Python, Jupyter Notebook, LaTeX, Git
Deep Learning: CNNs, U-Net family, ResNet, EfficientNetV2, Swin, CLIP, XLM-RoBERTa, attention mechanisms, ASPP, depthwise-separable convolutions
Methods: Medical image segmentation & classification, semi-supervised learning, multimodal learning, adversarial attacks/defenses, ablation design, multi-seed statistical evaluation
Training & Optimization: Mixed precision, focal/Tversky/boundary losses, class-imbalance handling, threshold tuning, test-time augmentation
Communication: IEEE-format papers, research posters, conference presentations; Bengali (native), English